

Public noncentral-endomorphism sampler: zero-run synthesis

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Goal: GOAL-SSI-2fa82a

Question: RQ-SSI-1946c9

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Result at a glance

[Independently verified] The round evaluated four proposed public noncentral-endomorphism sampler families against five independently owned joints. No family survived all five. F1 and F2 retain scoped accepted-output typing facts but break J3, J4, and J5; F3 remains inconclusive at J3 but breaks J4 and J5; F4 remains inconclusive at J3 and J5 but breaks J4. The round therefore reports zero surviving sampler families, zero PKE candidates, and zero KEM or key-exchange candidates. This is the copied, non-voted composition in [S7].

[Derived] “No surviving family” means only that none of the four frozen packets passed every joint. It is not a proof that public noncentral samplers do not exist, not an EndRing hardness result, and not a family-level impossibility theorem.

[Unmeasured] No scientific, implementation, formalizer, experiment, benchmark, or parameter-selection run was performed. Consequently this report contains no quantitative performance graph. The explanatory flow diagram is available as editable [Mermaid source](#) and as a rendered [vector SVG](#).

Evidence labels and decision loop

[Derived] This report uses four labels throughout:

- **[Proposed]** identifies a candidate mechanism, interface, or future gate.
- **[Derived]** identifies a consequence recomputed from the frozen contract or independently verified premises.
- **[Independently verified]** identifies a conclusion owned by a frozen independent review or the final composition.
- **[Unmeasured]** identifies a cost, probability, scaling, or performance quantity for which this zero-run round supplied no measurement.

[Independently verified] The user-directed research loop is explicit:

propose → independently evaluate → retain / repair / retire → ideate-next

[Derived] In this round, “retain” preserves the accepted-output algebra and the exact probability/cost identities that passed their owning joints; “repair” was considered under one reserved correction and skipped under the frozen admission rule; “retire” removes F1–F4 from the surviving set for this round without asserting universal impossibility; and “ideate-next” points to a new interface addressing adaptive OneEnd coverage. The complete flow and its status labels appear in Figure 1.

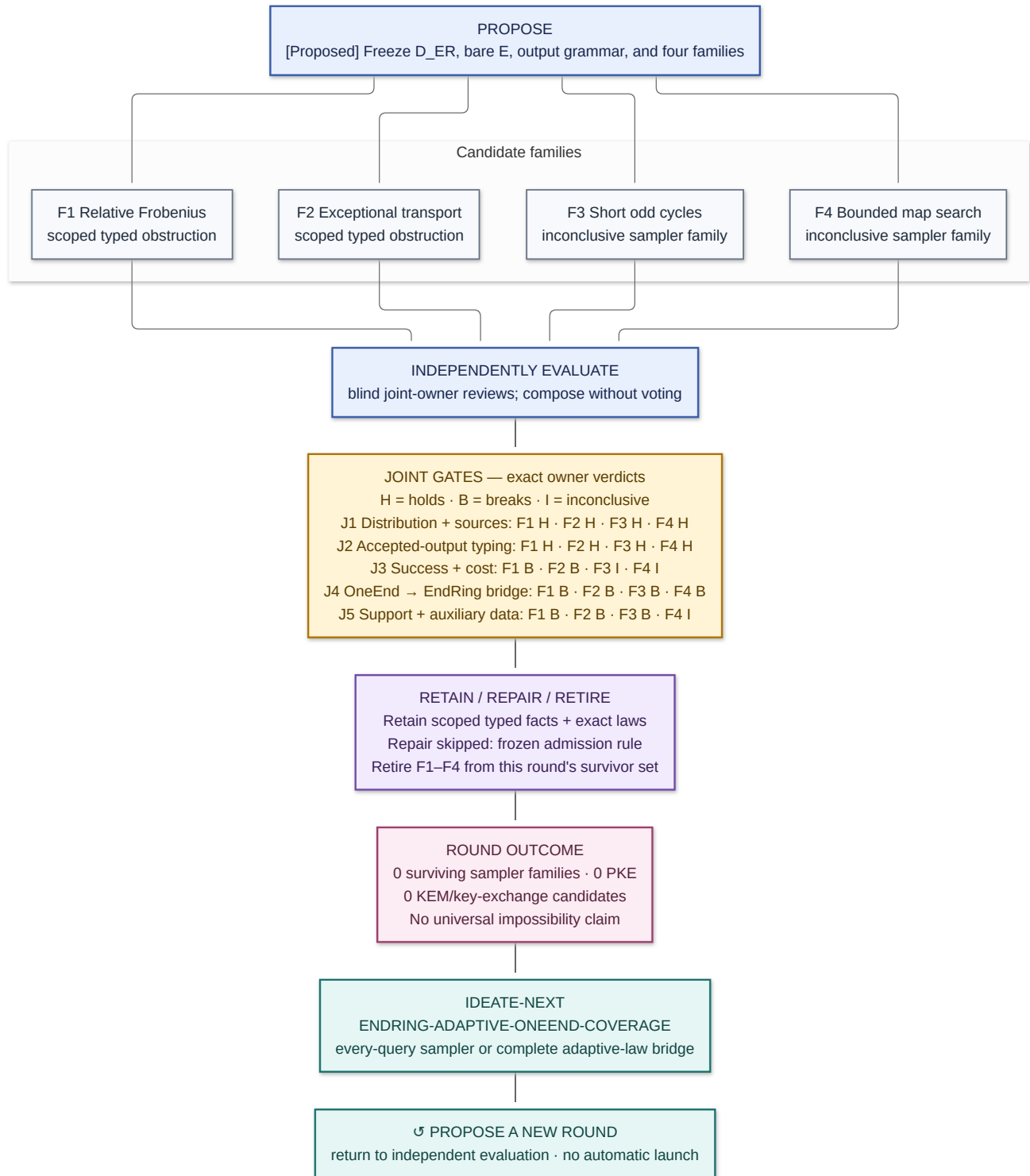


Figure 1. Candidate families feed independent J1–J5 gates, composition, the retain/repair/retire decision, and the next ideation loop.

Frozen question and scope

[Proposed] For public parameters $\Pi_\lambda = (p_\lambda, \mathbb{F}_{p_\lambda^2}, E_{0,\lambda}, \ell, L_\lambda, \text{Canon}, A_{\text{eval}})$, the challenge endpoint E is sampled by the campaign-defined exact nonbacktracking ℓ -isogeny walk from $E_{0,\lambda}$. Only canonical endpoint bytes are released. The walk, kernels, maps, orientation, ideals, tags, torsion labels, and endomorphism-ring witnesses are erased. The sampler receives only Π_λ , E , and independent coins [S1, S2].

[Proposed] The target interface is

$$\text{Samp}(\Pi_\lambda, E; \rho) \rightarrow \text{FAIL or } D_\alpha,$$

where an accepted descriptor is canonically serialized, efficiently evaluable, and publicly verifies as a non-scalar self-map $\alpha: E \rightarrow E$. It must provide one polynomial q and negligible ν such that

$$\Pr_{E \leftarrow D_{ER}} \left[\Pr_{\rho} [\text{Verify}(E, D_\alpha) = 1] \geq 1/q(\lambda) \right] \geq 1 - \nu(\lambda).$$

[Derived] This quantifier order separates bad-curve mass from coin failure on a fixed curve. Retrying with fresh coins must keep the same challenge E ; resampling E would change the experiment. The byte-level law includes FAIL, rejects, retries, aborts, descriptor size, validation, runtime, and memory [S1, S3].

[Independently verified] D_{ER} is a campaign-defined endpoint law on canonical bytes with path multiplicity retained. The inspected sources do not identify it exactly with a uniform or standard EndRing input law, and no applicable bridge to such a law was supplied [S2, S3].

Four proposed families

F1 — relative Frobenius and the conjugate curve

[Proposed] Find a separable $\phi: E \rightarrow E^{(p)}$, then compose it with relative Frobenius $F_{E^{(p)},p}: E^{(p)} \rightarrow E$. A deterministic public core exists on the locus where canonicalization supplies an isomorphism $c: E^{(p)} \rightarrow E$, using $\phi = c^{-1}$ [S1, S4].

[Independently verified] Conditional on a valid separable ϕ , the composition is a typed self-map of degree $p \deg \phi$ and inseparable degree p ; the nonsquare inseparable degree excludes scalars. A raw coordinatewise p -power map from E instead targets $E^{(p)}$ and is not an endomorphism of E unless the return leg is present [S2, S4, S5].

[Unmeasured] No general public FindConjugate, high- D_{ER} -mass bound, complete polynomial time/memory bound, or non-singleton honest-support result was established. The exact public core has publicly enumerable singleton support [S3, S7].

F2 — exceptional automorphism transport

[Proposed] Use a public nontrivial automorphism directly at $j = 0$ or 1728, or transport one to a general curve as $\widehat{\phi} \circ u \circ \phi: E \rightarrow E$ [S1, S4].

[Independently verified] Accepted direct exceptional automorphisms are degree-one maps distinct from ± 1 . For an accepted transported path, the quadratic identity has nonzero discriminant and rejects the identity-middle immediate backtrack. These are accepted-output facts only [S4, S5].

[Unmeasured] The two exceptional geometric classes do not supply a probability bound under D_{ER} . No general public FindExceptionalPath, complete polynomial cost, or hidden-support result was established; the direct route again has singleton public support [S2, S3, S7].

F3 — short nonbacktracking cycles

[Proposed] Enumerate bounded odd nonbacktracking ℓ -isogeny words, normalize every step, require exact canonical closure at E , and return the lexicographically first accepted descriptor [S1, S4].

[Independently verified] Every accepted odd closed word gives a typed self-map of degree ℓ^d , which is nonsquare for odd d ; immediate dual backtracks are excluded and would be scalar controls. The deterministic per-curve success law is exact [S3, S4].

[Unmeasured] F3-ODD-CYCLE-MASS and F3-PUBLIC-LINE-ENUM remain unproved, so neither high-mass success nor complete polynomial bit cost is established. The lexicographically first returned byte string is a publicly enumerable singleton whenever construction is polynomial [S3, S7].

F4 — bounded map or interpolation search

[Proposed] Sample a fixed bounded raw string, parse it as a canonical rational-map descriptor or deterministic interpolation transcript, compile to one explicit map, globally verify the curve equation and source/target, and reject scalars [S1, S4].

[Independently verified] Every accepted descriptor is a globally verified non-scalar $E \rightarrow E$ map. The raw domain, acceptance multiplicities, same-curve retry law, and one-call symbolic polynomial cost are exact under the frozen evaluator contract [S3, S4].

[Unmeasured] F4-BOUNDED-DESCRIPTOR-DENSITY is unproved. No positive support-entropy lower bound and no theorem excluding a different polynomial support enumerator was established. Thus J3 and J5 remain inconclusive, not positive [S3, S7].

Accepted-output typing is not construction or success

[Independently verified] J2 holds for all four families only under the quantifier “for every accepted non-FAIL descriptor.” It establishes the root type $E \rightarrow E$, field and composition compatibility, canonical syntax handling, and a public non-scalarly certificate [S4, S5].

[Derived] That statement has the logical form

$$\text{Accept} (E, D) \Rightarrow \llbracket D \rrbracket \in \text{End} (E) \setminus \mathbb{Z}.$$

It does **not** imply that an accepted descriptor exists for a general endpoint, that it can be constructed publicly from the bare curve, that the acceptance probability is inverse-polynomial on $1 - \nu$ of D_{ER} , or that retries, total cost, support, and scaling meet the sampler contract.

[Unmeasured] Evaluation through the frozen evaluator is specified, but no concrete operation count, runtime, memory measurement, implementation result, or parameter scaling was produced [S4, S5].

Independent joint matrix

[Independently verified] The following matrix reproduces the final composition exactly; the Coordinator copied owner verdicts without voting [S7].

Candidate	J1	J2	J3	J4	J5
F1-relative-frobenius	holds	holds	breaks	breaks	breaks
F2-exceptional-automorphism-transport	holds	holds	breaks	breaks	breaks

Candidate	J1	J2	J3	J4	J5
F3-short-cycle	holds	holds	inconclusive	breaks	breaks
F4-bounded-map-search	holds	holds	inconclusive	breaks	inconclusive

[Independently verified] Joint meanings are:

- **J1 — distribution and sources:** **holds** means the exact campaign law, bare-input declaration, representation boundary, and non-equivalence to a standard law are stated honestly. It does not prove useful-locus mass [S3].
- **J2 — type and non-scalarit:** **holds** is restricted to accepted outputs; it does not prove construction, existence, density, or scaling [S4, S5].
- **J3 — success and cost:** F1/F2 break because their submitted general routes and required success/cost parameters are absent; F3/F4 are inconclusive because named mass/density or cost obligations remain open [S6].
- **J4 — hard-problem implication:** **breaks** for the frozen interface across all families because the Page-Wesolowski adaptive oracle is not instantiated by a high-mass promise under one fixed distribution [S6, S7].
- **J5 — support and auxiliary data:** F1/F2/F3 break on publicly enumerable singleton returned support; F4 is inconclusive. Leaked path, orientation, ideal, tag, torsion, or ring data would change the bare-curve problem [S6].

Distributional bounded OneEnd versus the required adaptive oracle

[Independently verified] An accepted descriptor is a bounded OneEnd answer for that same curve. If a family actually satisfied J3, then for a good $E \sim D_{ER}$, capped same-curve repetition with $N = \lceil q(\lambda) \kappa \ln 2 \rceil$ would fail with probability at most $2^{-\kappa}$. Across D_{ER} , total failure would be at most $\nu(\lambda) + 2^{-\kappa}$ [S6, S7].

Property	Frozen sampler promise	Oracle needed by the cited PW reduction
Input coverage	Good set has mass at least $1 - \nu$ under one fixed D_{ER}	Every adaptively queried curve, or a proved bridge for every internal query law
Walk law	Fixed start, fixed length, nonbacktracking ℓ -walk	Source-generated 2-isogeny walks from the arbitrary EndRing input, with fixed and adaptive lengths
Failure	Capped wrapper can return FAIL ; unbounded retry may not terminate on bad curves	Each oracle call supplies a valid bounded non-scalar endomorphism
Cost condition	Expected polynomial only conditional on a good curve; capped cost includes residual failure	Uniform conditional expected bound through every adaptive transcript
Result	Distributional bounded OneEnd on the stated scope	Enough coverage to instantiate $\text{EndRing} \leq \text{bounded OneEnd}$

[Independently verified] The cited Page-Wesolowski direction is $\text{EndRing} \leq \text{bounded OneEnd}$: a bounded OneEnd oracle is used to solve EndRing. The reverse observation that an EndRing basis yields a non-scalar element is separate and must not be confused with that theorem [S2, S6].

[Independently verified] The earliest missing bridge is an upgrade to a total bounded OneEnd oracle for every adaptive query, or a complete bridge to all Page-Wesolowski internal query laws with reciprocal-success, termination, degree, representation, canonical-transport, and total-cost bounds. No such bridge is present. Therefore the frozen sampler-to-EndRing composition breaks; this is not an EndRing separation or hardness theorem [S6, S7].

Correction decision

[Independently verified] The reserved correction decision is reproduced exactly from [S7]:

```
reserved_task_id: TASK-20261009-1f7c04
user_authorization_received: true
queued: false
disposition: skipped_by_frozen_admission_rule
reason: >-
  No one missing source or typing repair can change a family disposition.
  F1 and F2 need a new public constructor plus mass/cost/support and oracle-
  coverage results. F3 needs independent mass/cost, support, and coverage
  repairs. F4 needs density, support, and coverage results. Changing D_ER or
  adding a path, orientation, ideal, tag, torsion data, or complete ring is
  forbidden by the frozen rule.
```

[Derived] No correction packet was admitted, so the initial matrix is the final matrix. The retained results are the scoped typed facts and exact laws; all four candidates are retired from this round's survivor set.

PKE, KEM, and key-exchange gates

[Derived] A public noncentral sampler would be only an upstream ingredient. The following gates remain distinct and none may be skipped:

Gate	Required evidence	Status after this round
Public sampler	One F1-F4-style family survives J1-J5, including construction, high-mass success, cost, and support	Not established: zero surviving families
Hard-problem bridge	Total adaptive bounded OneEnd coverage, or a complete bridge to every PW internal query law	Not established: J4 breaks
Public-key binding and correctness	A complete scheme syntax with key generation, encryption/encapsulation or exchange, decryption/decapsulation or agreement, and message/key recovery correctness	Not proposed or evaluated
Security game	Exact PKE/KEM/key-exchange adversary model and game, oracle access, assumptions, and a correctly directed reduction	Not proposed or evaluated
Concrete realization	Parameters, implementation, side conditions, validation, performance, and end-to-end costs	Unmeasured: zero implementation, parameter, experiment, and benchmark runs

[Independently verified] The composition establishes no secure PKE, no IND-CPA or IND-CCA result, no KEM, no authenticated or unauthenticated key exchange, no public-key binding, no message recovery construction, and no concrete parameters [S7].

Retain, retire, and ideate-next

[Derived] Retain:

- the exact D_{ER} and byte-law discipline;
- J2's accepted-output type/non-scalar invariants;
- F3/F4's exact per-curve counting and retry identities at their stated scope;
- the proof that a distributional promise does not automatically instantiate an adaptive always-answer oracle.

[Derived] Retire from this round's survivor set: F1, F2, F3, and F4 as submitted. F1/F2 need more than a single source or typing repair; F3/F4 remain scientifically open at named joints but still fail the all-joint admission rule. This retirement does not assert nonexistence.

[Proposed] Ideate-next under **ENDRING-ADAPTIVE-ONEEND-COVERAGE**: propose materially distinct interfaces that either give a uniformly polynomial public non-scalar-endomorphism sampler on every supersingular oracle input, or prove a complete bridge from a declared sampler law to every adaptive Page-Wesolowski internal query law. Each proposal must preserve exact success, total-cost, and honest-support accounting and must independently evaluate construction, typing, per-query success, adaptive termination, cost, support enumeration, and reduction quantifiers [S7].

[Proposed] The next ideation step is not an automatic launch and does not authorize a PKE/KEM/key-exchange or security claim. Hidden orientation, ideal, path, tag, torsion basis, or complete-ring data; challenge-curve resampling; and universal-impossibility inference remain excluded [S7].

Run and measurement accounting

[Independently verified] The frozen reviews and final composition report the following exact counts [S3, S4, S6, S7]:

Run class	Count	Evidence label
Scientific runs	0	independently verified
Implementation runs	0	independently verified
Formalizer runs	0	independently verified
Experiment runs	0	independently verified
Benchmark runs	0	independently verified
Parameter-selection runs	0	independently verified

[Unmeasured] There are no measured runtimes, operation counts, memory figures, success frequencies, uncertainty intervals, concrete security parameters, or end-to-end comparisons. A quantitative graph would therefore have no data to plot and is intentionally omitted. Source inspection, symbolic derivation, hashing, parsing, static validation, and document rendering are administrative operations, not scientific runs.

Limitations

[Derived] This report is a synthesis of the frozen methodology, source ledger, four independent review records, the J2 companion assessment, and the final composition. It does not re-open producer packets or primary-source bytes and does not extend any frozen reviewer's authority.

[Independently verified] The results do not establish an EndRing solver, EndRing hardness, complete-ring reconstruction, novelty, a deployable primitive, a parameter set, or impossibility of public noncentral samplers. J6 documentation review occurs only after these report/visual artifacts exist and is not claimed here [S7].

Source index

[Derived] All scientific claims above are traceable to these frozen repository artifacts. Hashes are the frozen values recorded by the review or composition; [S1] is the unchanged methodology hash recorded by J4.

ID	Frozen repository source	Role	SHA-256
S1	methodology.md	Frozen question, families, controls, and zero-run boundary	0c2cbca1fd2e78b2d4bbfe138d63e92fe6643070f3d5401ff1a308d392e88998
S2	source-ledger.yaml	Primary-source provenance and transfer limits	d7a7e0063115037ac77302a8e748d899fea6e6ec0732660efedd6fc56f6336fa9
S3	J1 review	Exact distribution, bare input, and source boundary	9d0807b0b05b0598001fbbb80533e0eba548f5fb7028f99ab2663a12a10b4c2e
S4	J2 review	Accepted-output typing, serialization, and non-scalarly	3f11486bf45d2b77f3f03850cad64a097b1fb894d5c4c95997d38ed8e6bac111
S5	J2 transfer companion	Construction/existence/preservation/cost separation; draft companion, not a ledger record or certificate	d87296a557b2147cbd0285d06bb2ecee9a5fcb5691eb37483896ebc9cbfc3d0c
S6	J3/J5 review and J4 review	Success/cost/support and hard-problem implication	e9b9478b7a26cee0512e81194f2a0efa55871c6486447d1a00cf8ad657fcd4c; 9f99aa52b39d29aca1328e1dbec9b9840a5a1a4487878cdf9845ddca38663924
S7	final-composition.yaml	Exact matrix, correction decision, outcomes, and next action	a16c3cc8d6302723b839eeaa310a308939870933cae53f0d00f7a90da78f7533